

Computer vision

Vanda Benešová

Generative NN for visual data synthesis

Variational autoencoder (VAE)

Variational autoencoder (VAE) is generative model used to generate new image data in the form of variations of the input data they're trained on.

Pros:

Stable Training, Structured Latent Space, Interpretable & Controllable.
Probabilistic Framework: Can estimate data likelihood and are also good for tasks like anomaly detection or denoising.

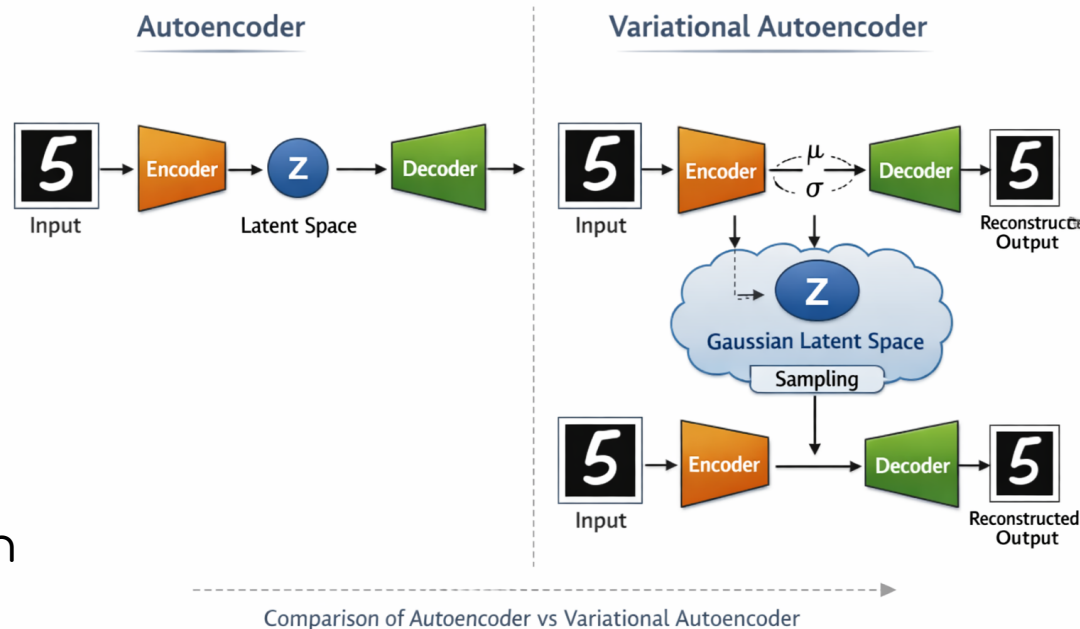
Cons:

Blurry Samples, Loss of Detail, Hyperparameter Sensitivity

Variational Autoencoder (VAE)

A Variational Autoencoder (VAE) works by compressing data (like images) into a probabilistic, continuous latent space (mean and variance) and then reconstructing it, allowing it to generate new, similar data by sampling from this organized space.

Unlike most autoencoders, which encode a single vector of discrete latent variables, VAEs are probabilistic models.

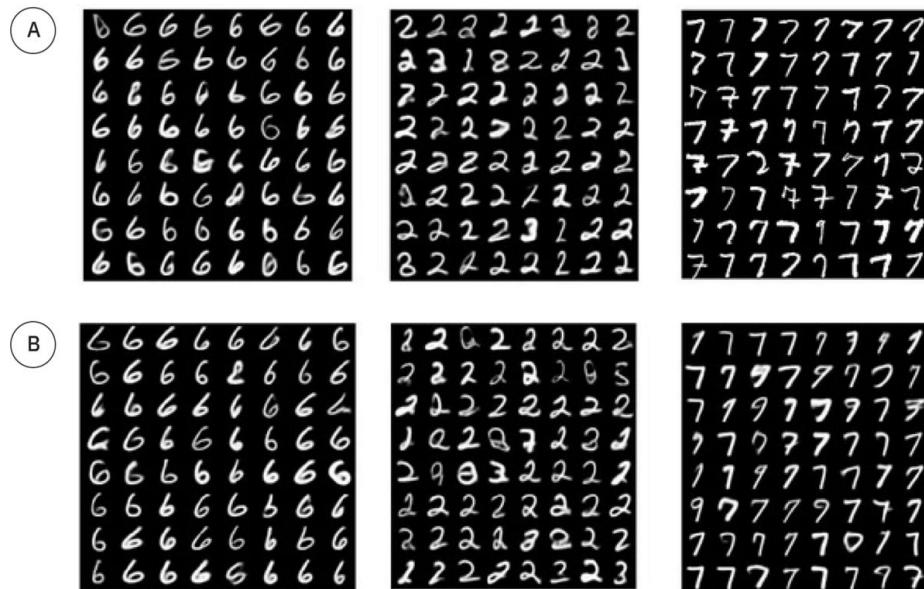


Variational autoencoder (VAE)

- Learns a continuous latent representation
- Enables controlled and meaningful data generation
- Widely used in image synthesis, anomaly detection, and representation learning

The decoder takes the random sample from the latent space and tries to reconstruct the original input.

Since the encoder gives a range, the decoder can produce new data that is similar but not identical to what it has seen



A: handwritten digits from the MNIST data set;
B original samples output by a VAE trained on those MNIST digits.

Source: Mak, Hugo & Han, Runze & Yin, Hoover. (2023)

Generative adversarial network (GAN)

A GAN (Generative Adversarial Network) is a deep learning model using two competing neural networks: a Generator and a Discriminator to create new, realistic data (like images) from a training set.

Pros:

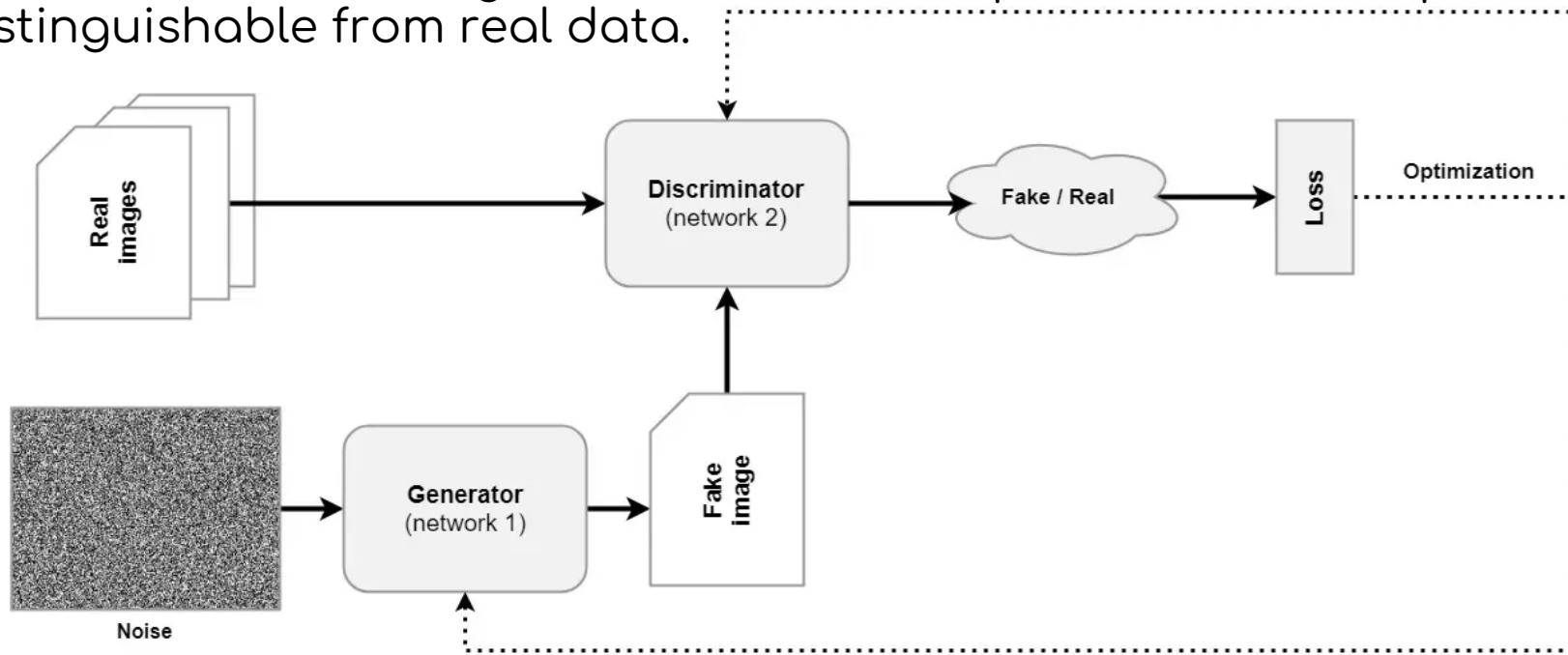
High-Quality Output

Cons:

Training instability, Mode collapse and Mode Dropping, small samples diversity

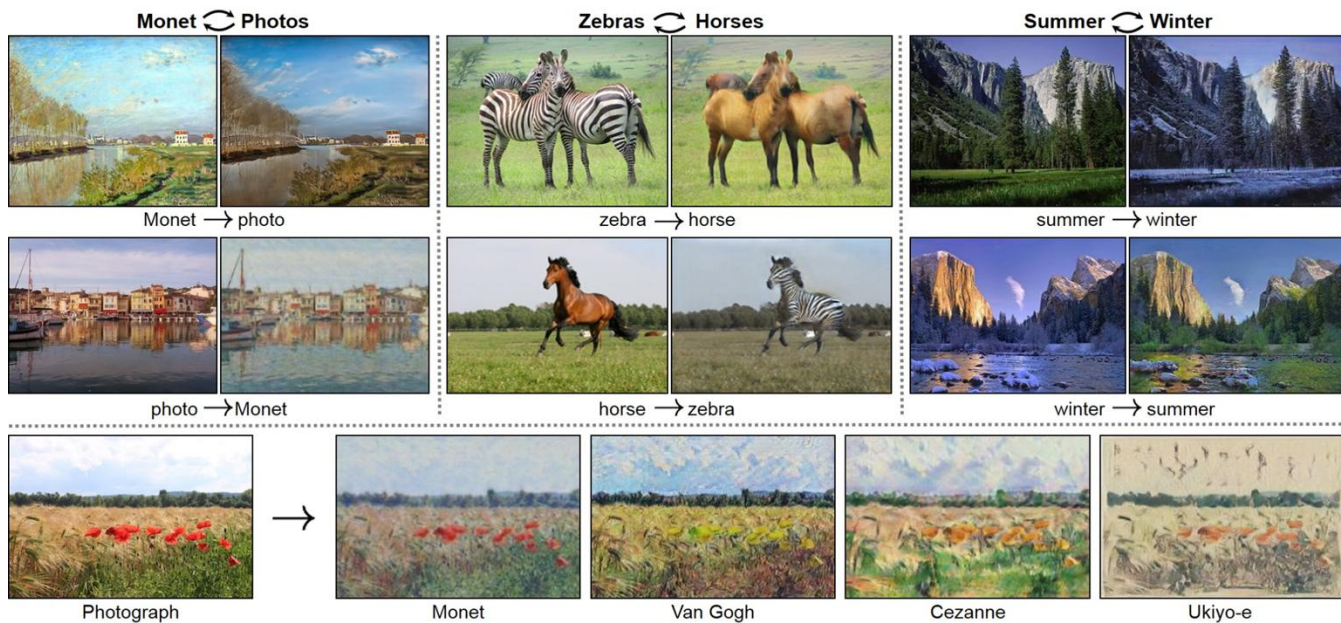
Generative adversarial network (GAN)

The Generator makes fake data, while the Discriminator tries to distinguish the fakes, forcing the Generator to improve until its outputs are indistinguishable from real data.



GAN Modification: CycleGAN

Image-to-image translation



<https://junyanz.github.io/CycleGAN/>

GAN Modification: CycleGAN

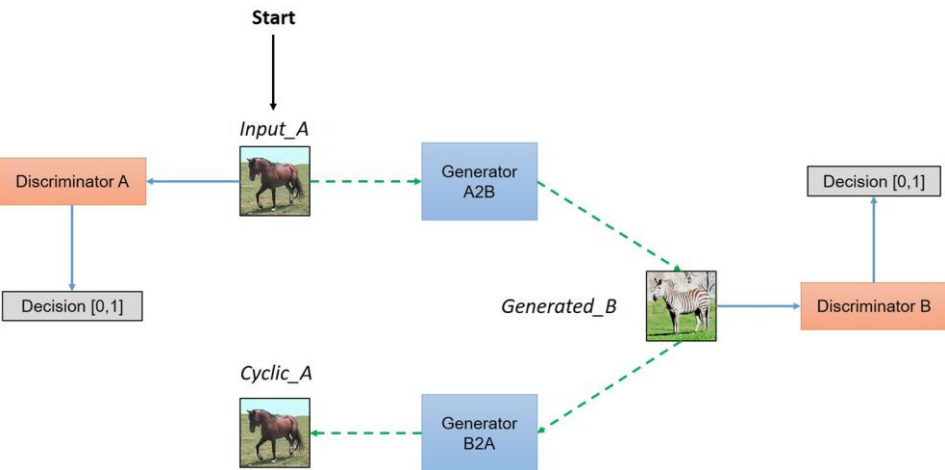
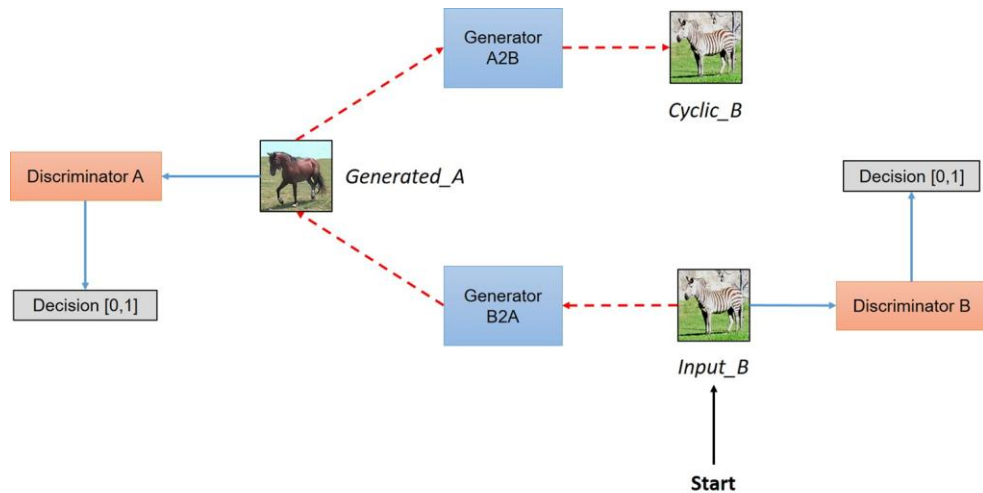


Image-to-image translation



GAN modification StyleGAN :

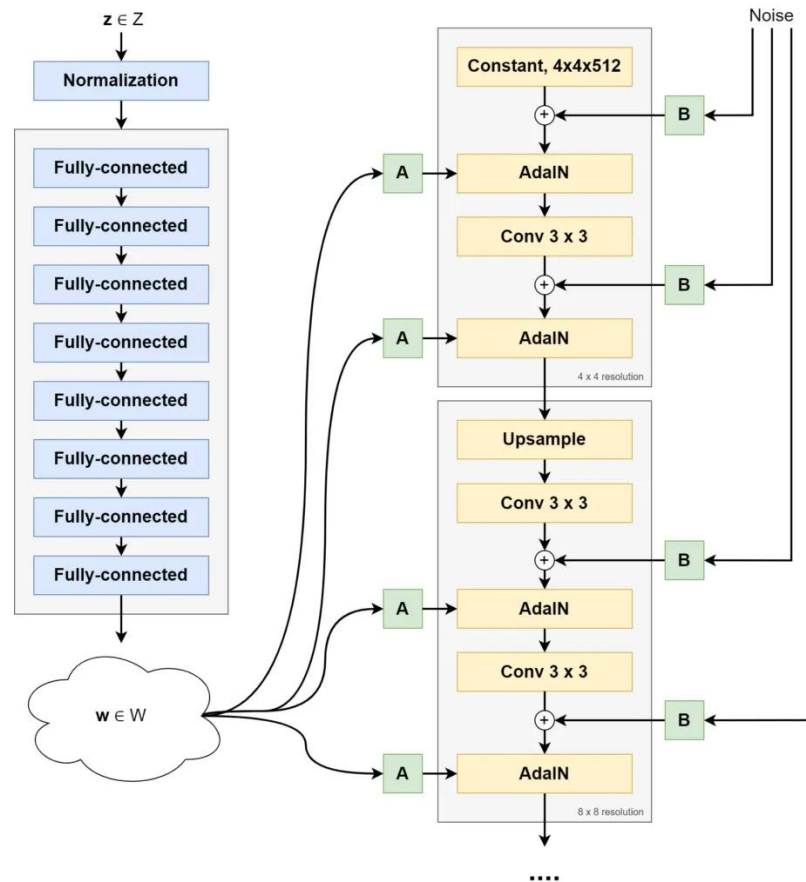
demo <https://youtu.be/kSLJriaOumA>

There are two vertical blocks involved:

The mapping network, called f , is visible on the left.

It maps a (normalized) latent vector $z \in Z$ into another vector w from an intermediate latent space, called W .

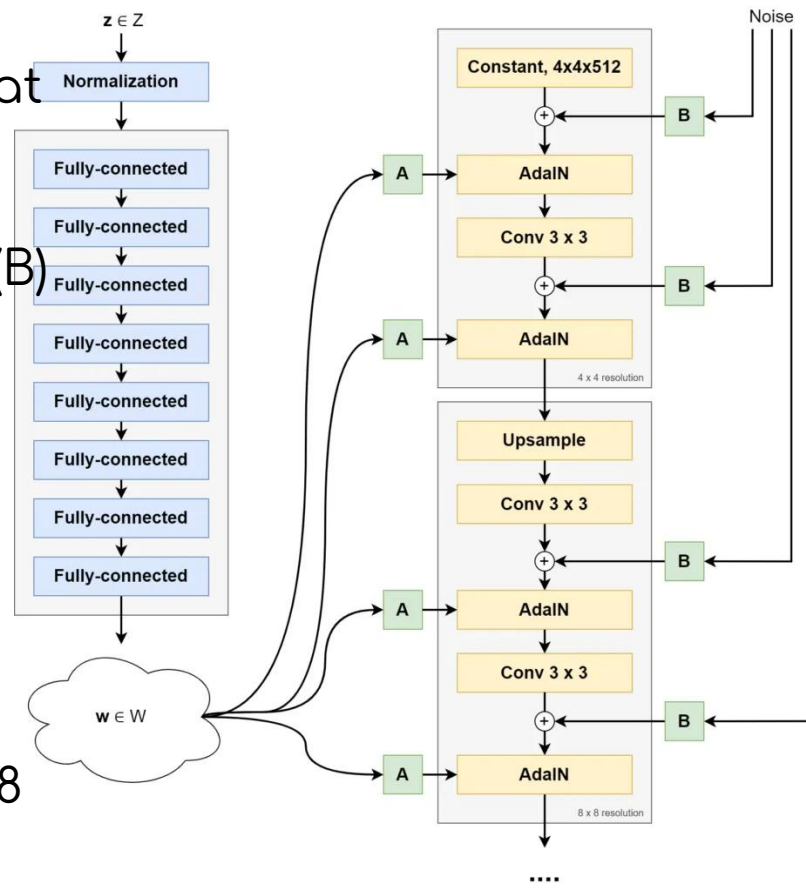
This mapping network is a simple set of fully-connected feedforward layers.



GAN modification StyleGAN :

The synthesis network, called g and visible on the right, uses w to generate a “style” that controls the image synthesis process.

It begins with a Constant, $4 \times 4 \times 512$ dimensional vector. Scaled noise samples (B) are generated and added to this Constant tensor. Subsequently, the style (A) is added via Adaptive Instance Normalization operations, after which a convolution operation is applied. This is followed by another noise addition and AdaIN-based styling operation. We then arrive at an image at a 4×4 pixel resolution. In the next block, the image is upsampled, and the same is performed again, arriving at an 8×8 pixel resolution. This is repeated until the image is 1024×1024 pixels.



Diffusion models

Diffusion models are generative models that “diffuse” training data with random noise, then learn to reverse the diffusion process to output new images.

Pros:

High-Quality Image Synthesis, Diverse Sample Generation, Stable Training Objective

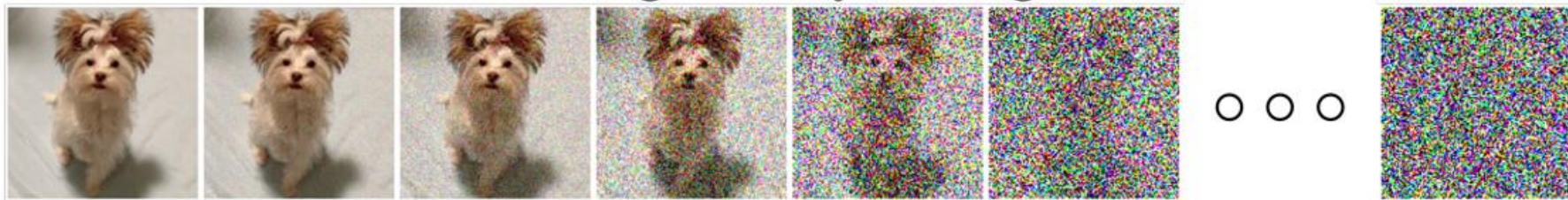
Cons:

High Computational Cost, Slow Generation Speed

Diffusion models

The neural network is trained on a dataset of real images. The training process teaches the neural network how to add detail to noise images in a way that is consistent with the training data. Once the neural network is trained, it can be used to generate realistic images.

Data ——— Destructing data by adding noise ———→ Noise



Data ←—— Generating samples by denoising ——— Noise

Diffusion models

- Forward Diffusion Process (Training):

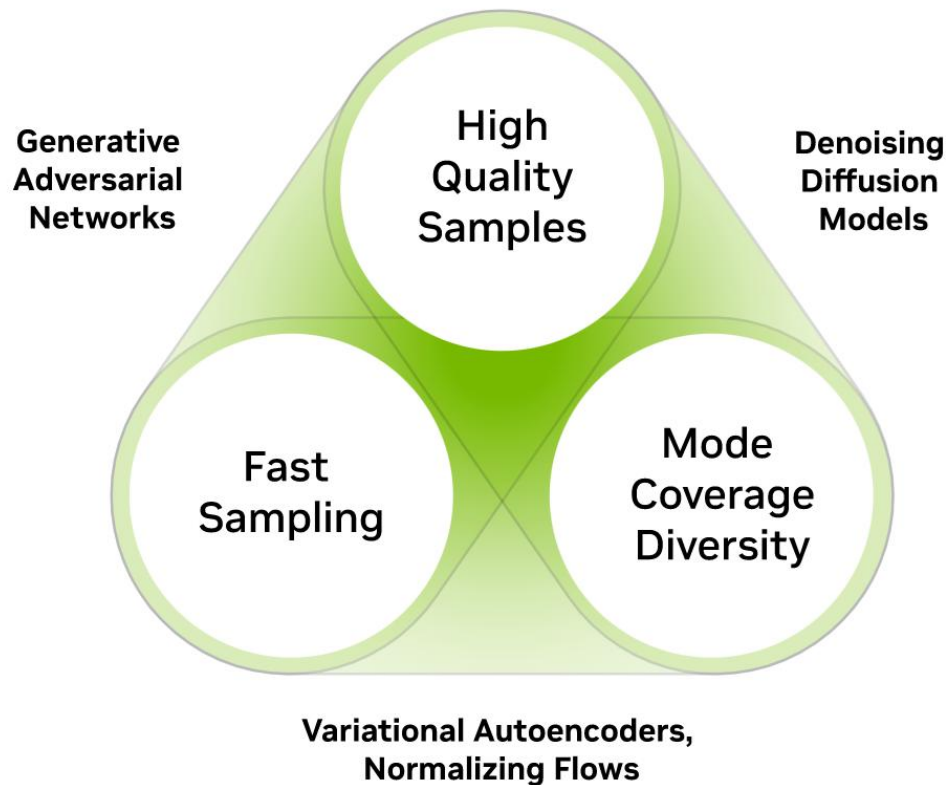
The model takes a clean image and systematically adds small amounts of Gaussian noise to it over many steps, eventually transforming the image into pure, unstructured noise.

- Reverse Diffusion Process (Generation):

This is the crucial part where a neural network (often a **U-Net** or a Transformer) is trained to reverse the forward process. It learns to predict and subtract the noise at each step, essentially learning how to "denoise" the data.

Generative AI models - vision tasks

Trilemma

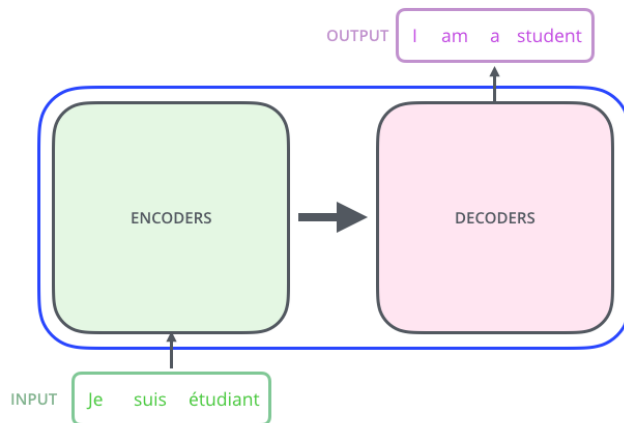


Vision Transformer (ViT)

Transformer model

Deep learning architecture, introduced by Google in 2017, that revolutionized AI by processing entire sequences (like text) simultaneously, unlike older models that processed them word-by-word.

Its core innovation is the **self-attention mechanism**, allowing it to weigh the importance of different words in a sequence, capturing long-range context efficiently.



<https://www.youtube.com/watch?v=GvezxUdLrEk>

Transformer model

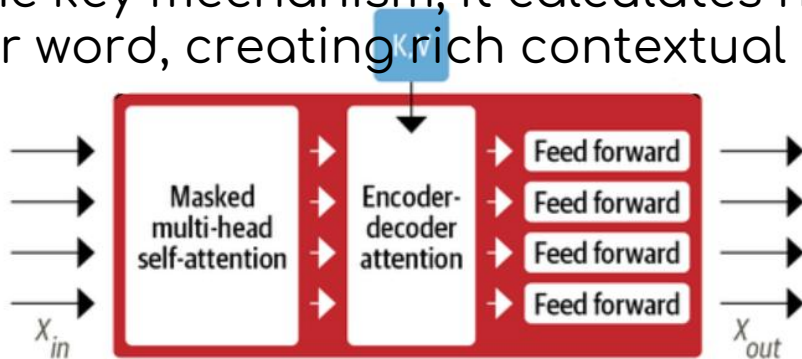
Components of Encoder-only Transformer Architecture

Multiple Encoder Layers: Typically contains several layers of encoders, each comprising multi-head self-attention and feed-forward neural networks. The picture below represents the same.

Positional Embedding: Adds positional information to the input tokens to understand the sequence of words in a sentence.

Self-Attention Mechanism: The key mechanism; it calculates how much each word relates to every other word, creating rich contextual understanding.

This enables the model to consider other words in the input when encoding a particular word.



Vision Transformer (ViT)

1. Split an image into patches
2. Flatten the patches
3. Produce lower-dimensional linear embeddings from the flattened patches
4. Add positional embeddings
5. Feed the sequence as an input to a standard transformer encoder
6. Pretrain the model with image labels (fully supervised on a huge dataset)
7. Finetune on the downstream dataset for image classification



Vision Transformer (ViT) Attention

An attention mechanism is a machine learning technique that directs deep learning models to prioritize (or attend to) the most relevant parts of input data.

Attention = $f(Q, K, V)$

Q: The query vector represents the information a given token is seeking.

K: The key vectors represent the information that each token contains. Alignment between query and key is used to compute attention weights.

V: The value (or value vector) applies the attention-weighted information from the key vectors. Contributions from keys that are strongly aligned with a query are weighted more heavily; contributions from keys that are not relevant to a query will be weighted closer to zero.

Vision Transformer (ViT)

Self-Attention vs Cross-Attention

Self-attention is when the query, key, and value all come from the same input sequence.

Cross-attention is when the query comes from one source, and the key and value come from another.

Feature	Self-Attention	Cross-Attention
Input Source	Single input	Two different inputs
Purpose	Capture context within a sequence	Fuse information across sequences or modalities
Common Use	Encoders, Decoders	Decoder-to-Encoder, Multimodal alignment
Parameters	Shared weights for Q, K, V	Separate projections for Q (from X) and K, V (from Y)
Example	BERT, GPT	Transformers for translation, CLIP, Flamingo

LLM
Examples

Swin Transformer

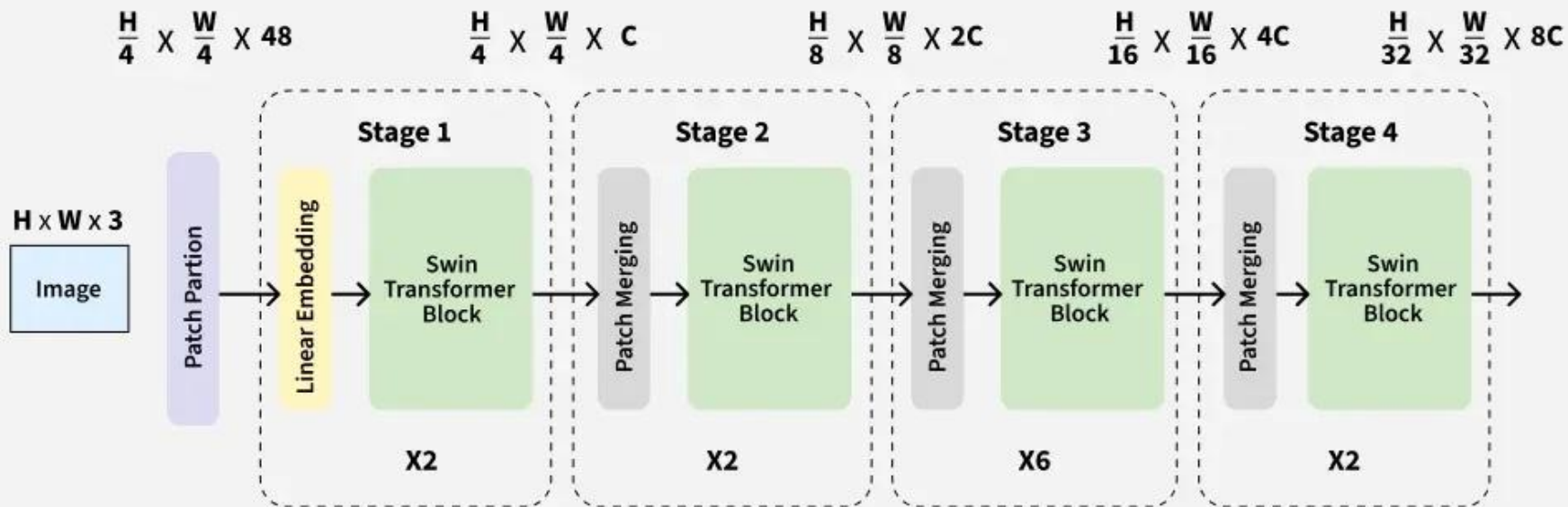
Swin Transformer

The Swin Transformer (Shifted Window Transformer) is a type of vision transformer model that processes images by dividing them into small, non-overlapping windows and computes self-attention within these localized regions.

Unlike standard vision transformers which use global attention, Swin Transformer introduces a "shifted window" technique.

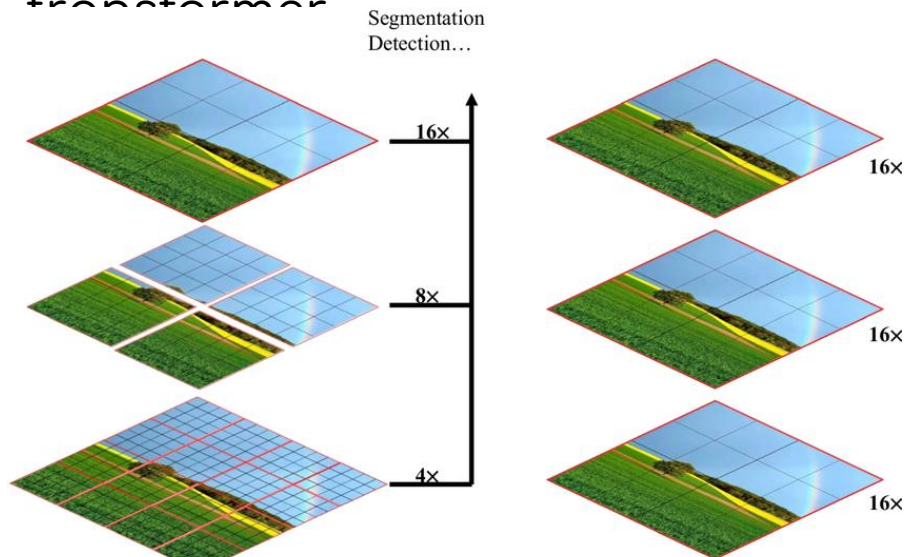
This allows neighboring windows to interact with each other in subsequent layers, efficiently capturing both local and global features in an image.

Swin Transformer



Swin Transformer

Patch Splitting: The input image is divided into fixed-size patches like putting a grid over image and each square represent a patch. Each patch is then embedded into a feature vector to form input for the transformer



Comparison between swin transformer and Vision Transformer (ViT)

Swin Transformer

Window-Based Self-Attention: Instead of computing attention globally the model computes attention within local windows. These windows act as small focused regions capturing fine features while keeping computation manageable. Self-attention is applied within the window and captures local features.

Shifted Windows for Cross-Region Interaction: The shifted window mechanism solve limitation of local windows attention and capture global context of image. This shifted window shifts the position of the windows by a small value and hence overlapping regions with next layer. This ensure cross-window communication and improve models ability to capture global context.

Segment Anything

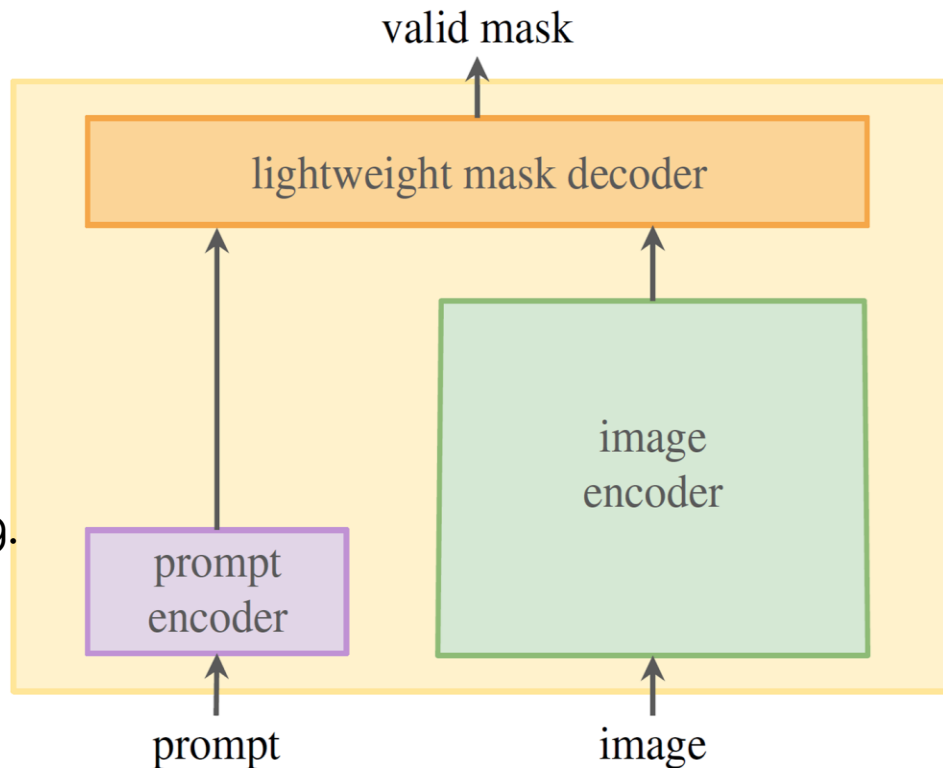
SAM 1 Segment Anything

META, Open Source

The project includes the Segment Anything Model (SAM) and the SA-1B dataset.

- SAM is designed to perform promptable segmentation, allowing it to adapt to various tasks without additional training.

- [SAM 1](#)



Model: Segment Anything Model (SAM)

SAM 1 Dataset

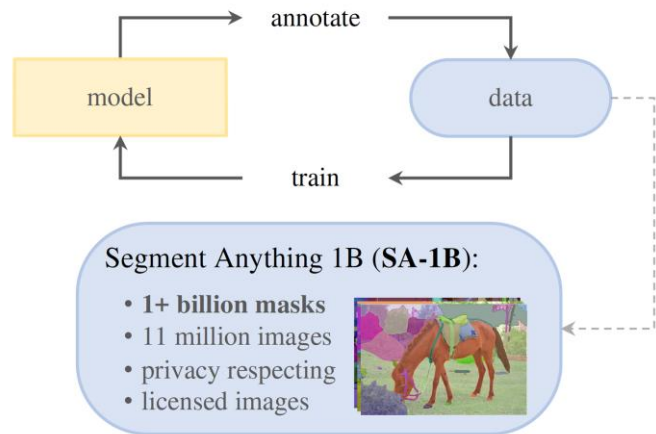
1 billion masks derived from 11 million licensed images

Assisted-manual stage. In the first stage, resembling classic interactive segmentation, a team of professional annotators labeled masks by clicking foreground / background object points using a browser-based interactive segmentation tool powered by SAM.

Semi-automatic stage. In this stage, we aimed to increase the diversity of masks in order to improve our model's ability to segment anything. To focus annotators on less prominent objects, we first automatically detected confident masks. Then we presented annotators with images prefilled with these masks and asked them to annotate any additional unannotated objects.

Fully automatic stage. Two major enhancements to our model:

- First, we had collected enough masks to greatly improve the model, including the diverse masks from the previous stage.
- Second, by this stage we had developed model, which allowed us to predict valid masks even in ambiguous cases. Specifically, we prompted the model with a 32×32 regular grid of points and for each point predicted a set of masks that may correspond to valid objects.

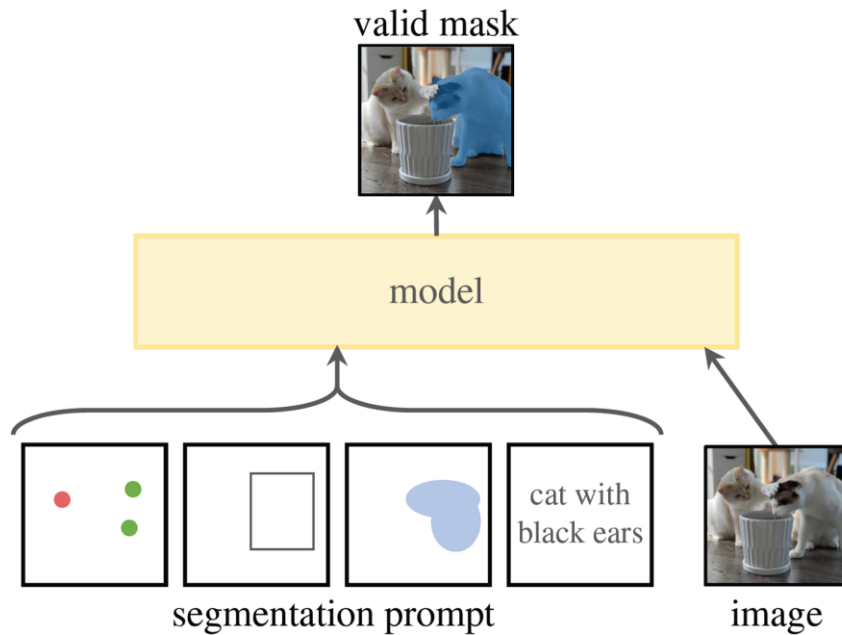


(c) **Data:** data engine (top) & dataset (bottom)

SAM 1

META, Open Source

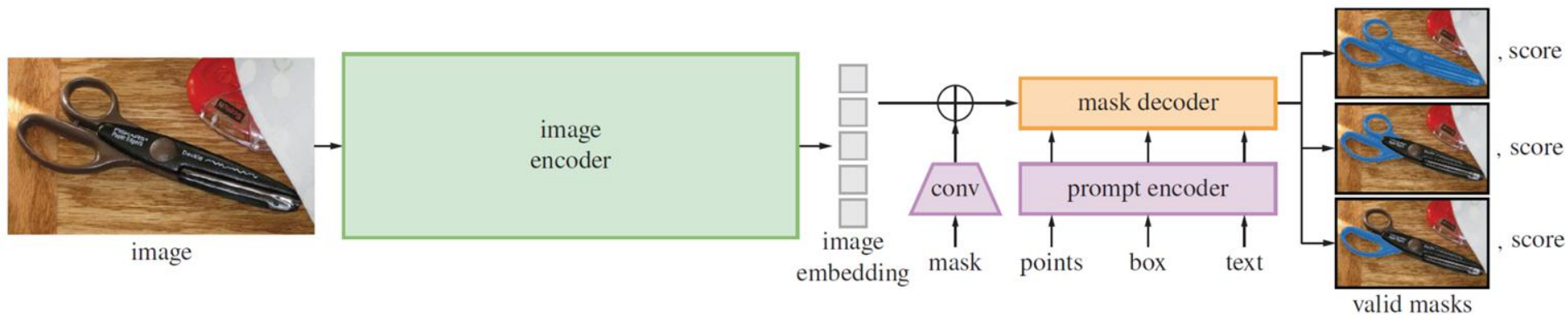
Promptable segmentation:
point, box, and mask prompts, and
also free-form text prompts.



(a) **Task:** promptable segmentation

<https://ai.meta.com/blog/segment-anything-foundation-model-image-segmentation/>

Segment Anything Model overview



A heavyweight image encoder outputs an image embedding that can then be efficiently queried by a variety of input prompts to produce object masks at amortized real-time speed.

For ambiguous prompts corresponding to more than one object, SAM can output multiple valid masks and associated confidence scores.

SAM 2

SAM 2 can segment any object in any video or image- even for objects and visual domains it has not seen previously, enabling a diverse range of use cases without custom adaptation.

SAM 2 has many potential real-world applications.

For example, the outputs of SAM 2 can be used with a generative video model to create new video effects and unlock new creative applications. SAM 2 could also aid in faster annotation tools for visual data to build better computer vision systems.

SAM 3

SAM 3 supports:

a variety of prompt modalities, including both

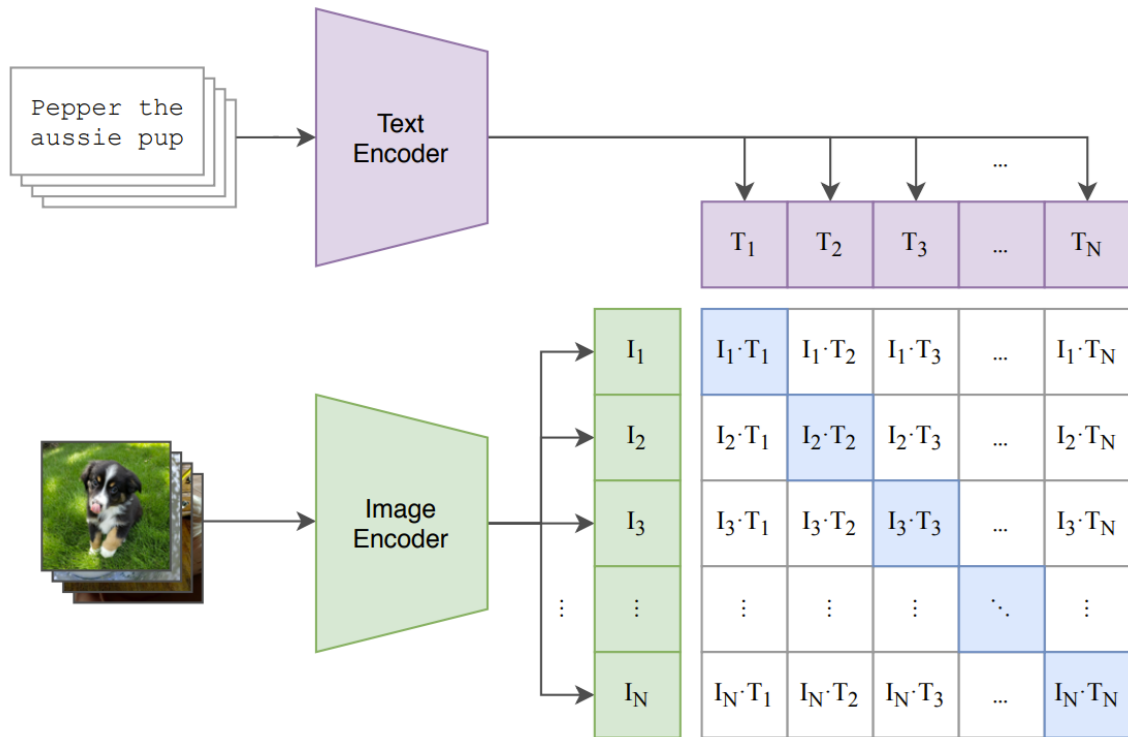
- concept prompts such as simple noun
- and image exemplars,
- as well as visual prompts, such as masks, boxes, and points, which were introduced in [SAM 1](#) and [SAM 2](#).

CLIP

CLIP = Contrastive Language-Image Pretraining

OpenAI, 2021

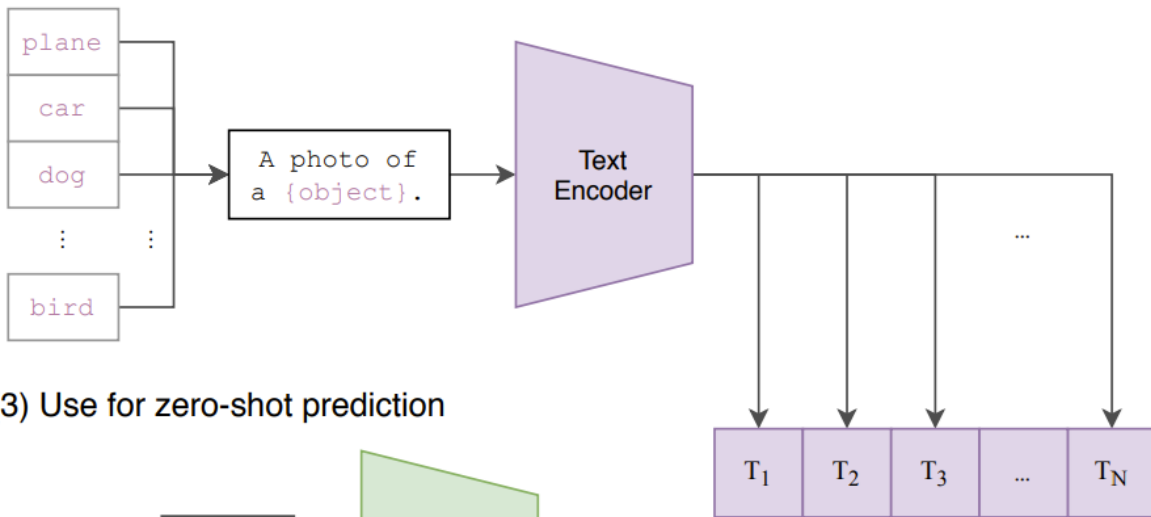
(1) Contrastive Pretraining



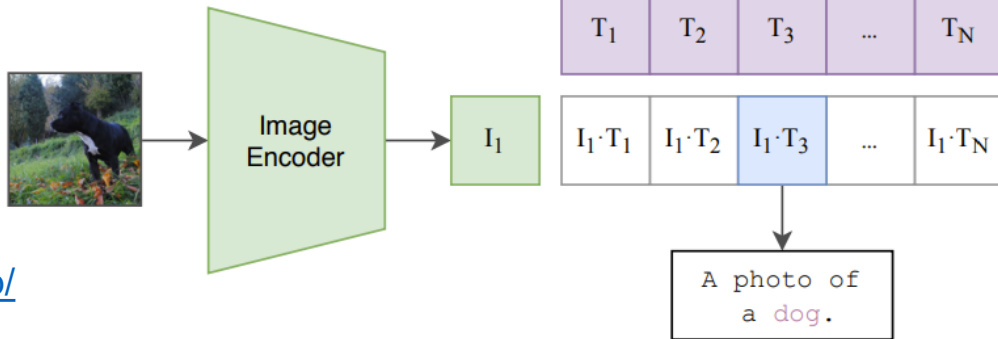
CLIP - “zero-shot” classification

OpenAI, 2021

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

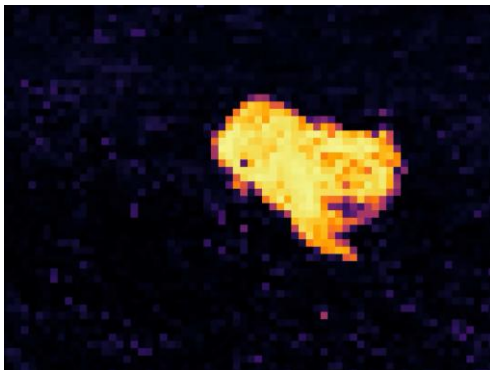


DINO

DINO = Self-Distillation with NO Labels

DINOv1

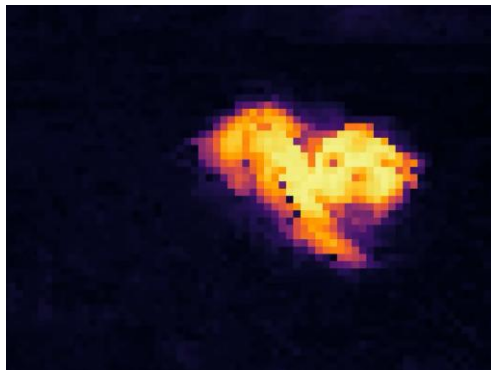
Caron et al. 2021



Proof-of-concept, with 80M-parameter models trained on 1M images.

DINOv2

Quab et al. 2024



Scaling of a SSL algorithm. 1B-parameter models trained on 142M images.

DINOv3

Siméoni et al. 2025



An order of magnitude larger training compared to v2, with particular focus on dense features.

DINO = Self-Distillation with NO Labels

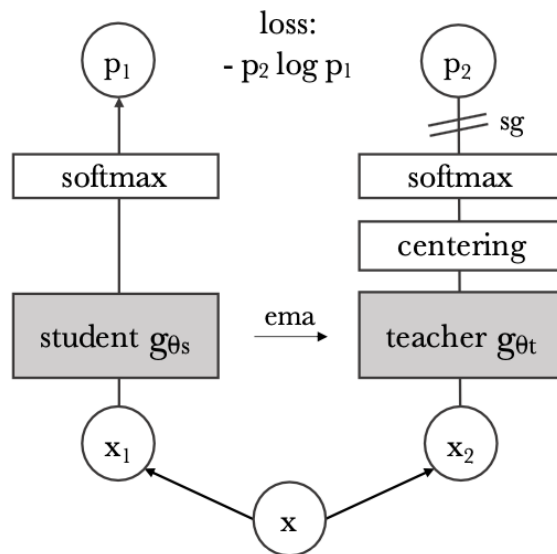
Meta AI, Caron et al. 2021



Cosine similarity maps obtained with DINOv3 output features between the patches marked with a red cross and all other patches.

DINO = Self-Distillation with NO Labels

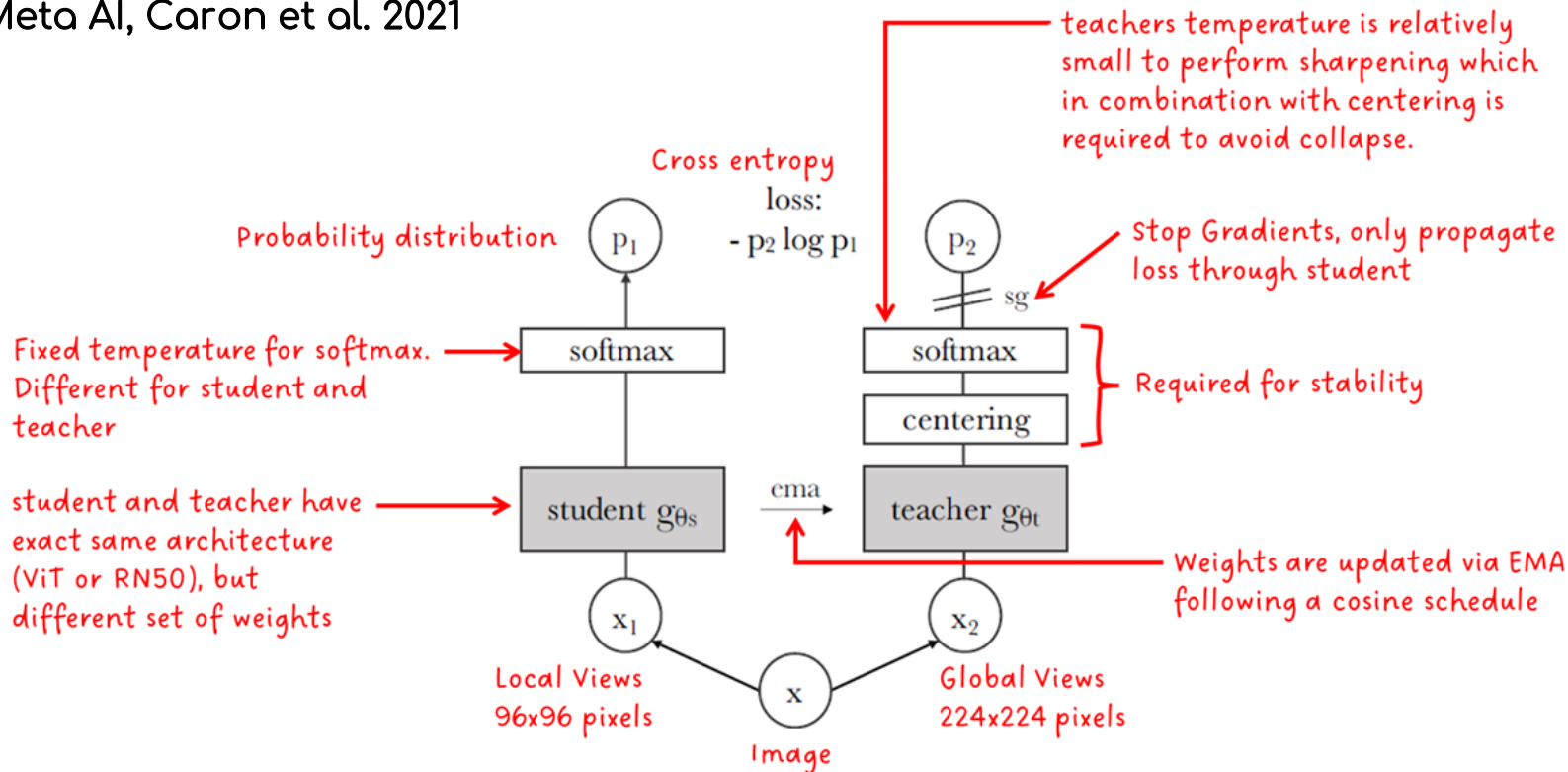
Meta AI, Caron et al. 2021



*DINOv1 architecture

DINO = Self-Distillation with NO Labels

Meta AI, Caron et al. 2021



*DINOv1 architecture

VGGT: Visual Geometry Grounded Transformer

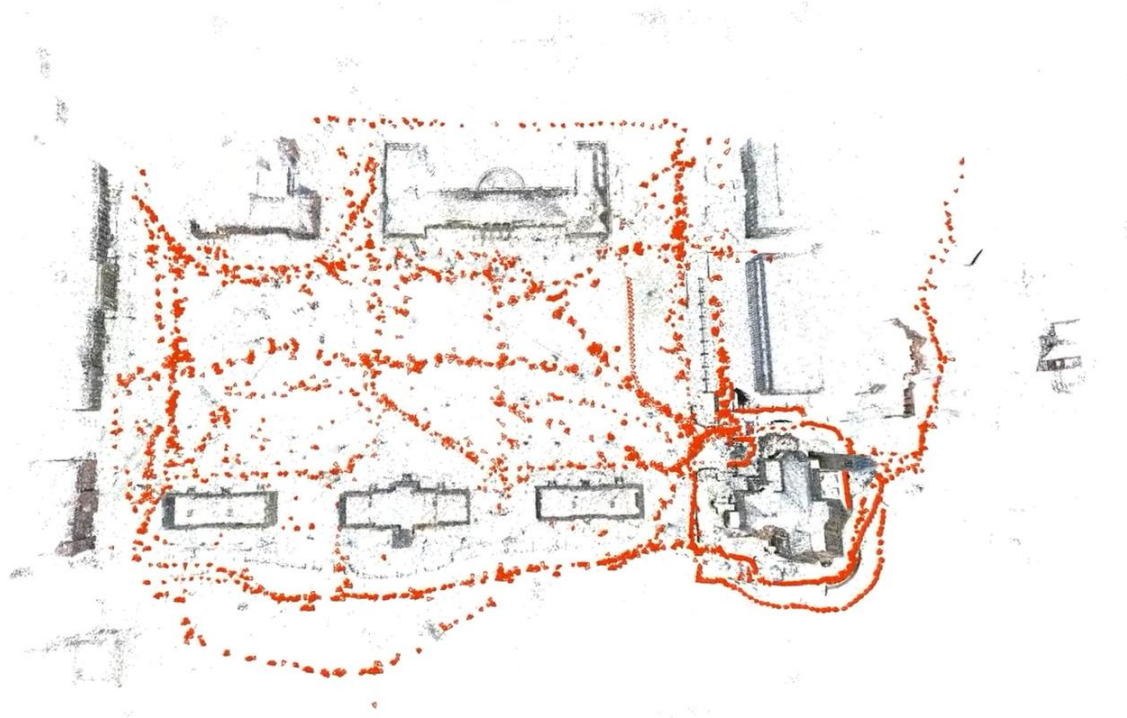
Jianyuan Wang^{1,2}, Minghao Chen^{1,2}, Nikita Karaev^{1,2},
Andrea Vedaldi^{1,2}, Christian Rupprecht¹, David Novotny²
¹Visual Geometry Group, University of Oxford, ²Meta AI

CVPR 2025

Published to arXiv: 3/14/25

*slides adapted from [Jianyuan](#)

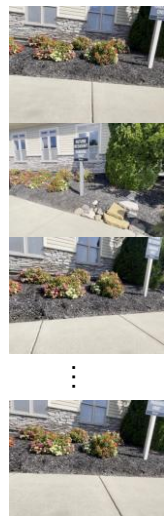
3D Reconstruction



COLMAP, Schönberger et al. 2016

3D Reconstruction - VGGT

Images



Neural
Network



3D Reconstruction
Camera parameters, Depth maps,
Point maps and Point correspondences

Main contributions

Paradigm shift

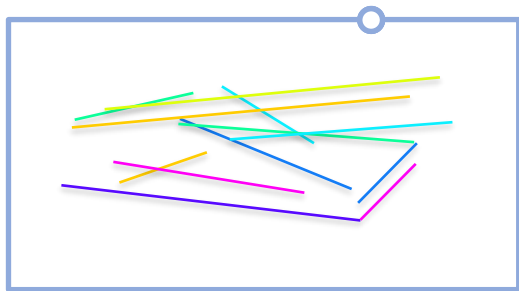
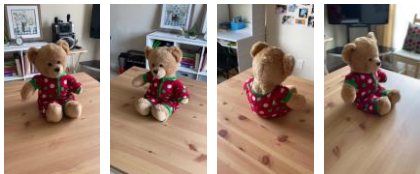
- Previous paradigms are dominantly pairwise approaches
- Only two views are used during training
- Test-time fusion usually needed

Foundation model for 3D tasks

- Since its release many follow up work has been done
- Most of them treat VGGT as a backbone

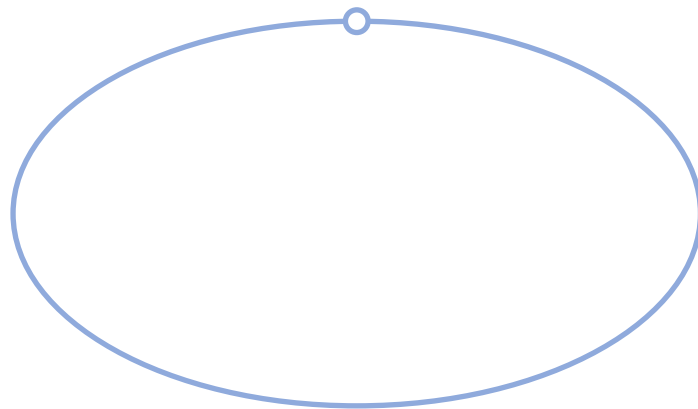
COLMAP vs DUS3R

COLMAP



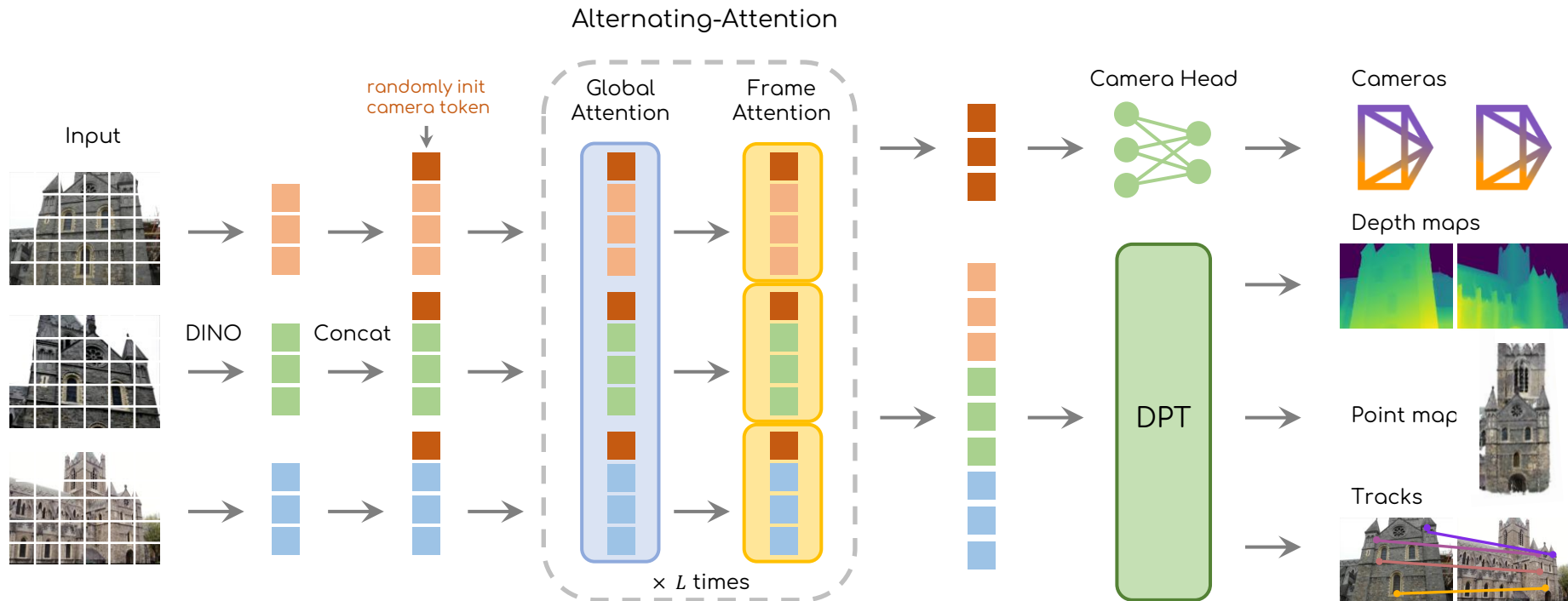
Bundle Adjustment

DUS3R



Global Alignment

Method: VGG Transformer



Method: VGG Transformer

Aggregator based on DINOv2 (Quab et al. 2024) – ViT L/14 with 4 register tokens

- Linear projection (tokenization) weights initialized with pretrained DINOv2 weights
- Uses RoPE (Su et al. 2023) as in DINOv3 for patch PE

Camera head

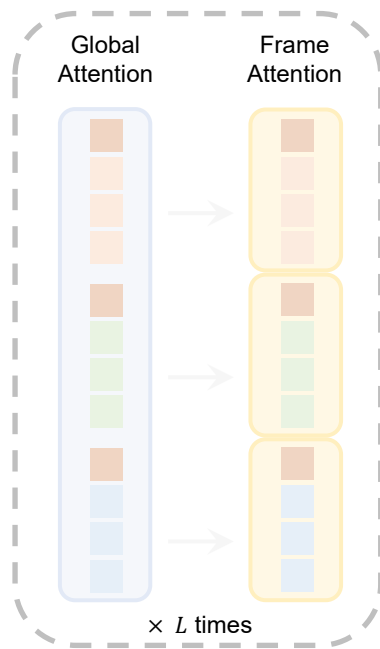
- Uses four additional self-attention layers followed by a linear layer

DPT head (Ranftl et al. 2021)

- For dense predictions of Depth map, Point Map and Tracking features

Why Alternating-Attention?

- Global Attention
 - Ensures scene-level coherence
- Frame-wise Attention
 - Eliminates **frame index embedding**
 - For permutation equivariance
 - For flexible input length



Why Alternating-Attention?

Frame 0



$\text{Embed}(0)$

Frame 1



$\text{Embed}(1)$

Frame 2



$\text{Embed}(2)$



$\neq$

Model(



)

Model(



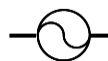
)

Not permutation equivariant

Why Alternating-Attention?

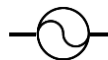
Frame 0



 $Embed(0)$

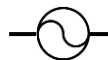
Frame 1



 $Embed(1)$

Frame 2



 $Embed(2)$

⋮

Frame 842

But model never sees $Embed(842)$ during training

Why Alternating-Attention?

Frame 0



 $Embed(0)$

Frame 1

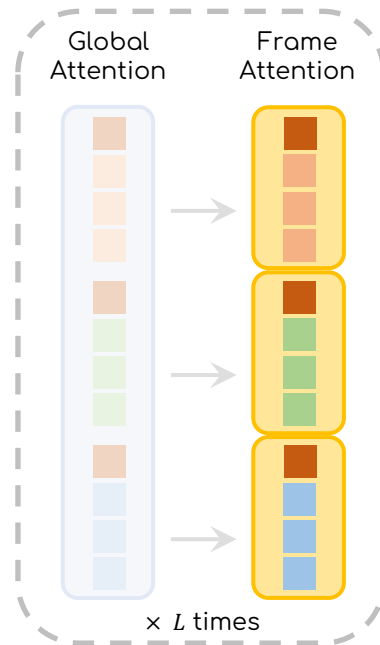


 $Embed(1)$

Frame 2



 $Embed(2)$

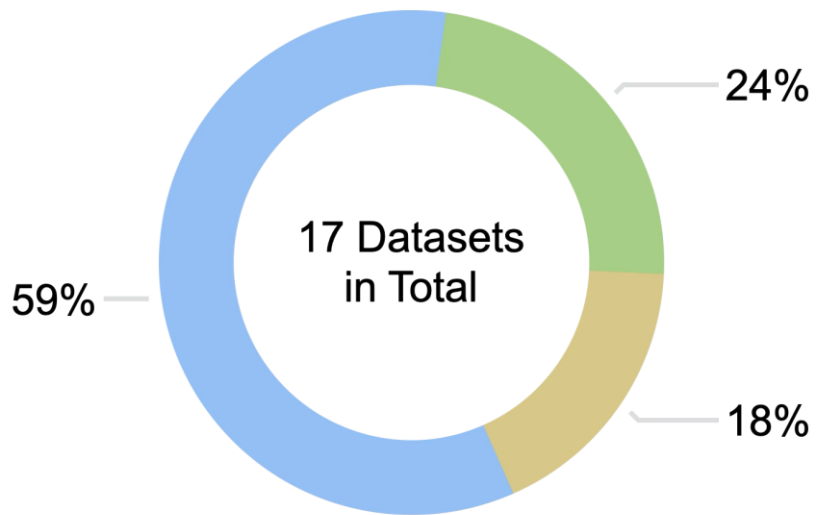


Replaces frame index embedding by Frame-wise Attention

Training and Data

□ Training:
2 to 24 frames

□ Inference:
1 to 300+ frames



● Synthetic

● Captured

● SfM-Annotated



Accuracy



Generalization

Results

32 Views



Zero-shot Monocular Depth Estimation

Single View



General AI - Yes or No?

AGI = Artificial General Intelligence

- *What is AGI?* <https://arxiv.org/pdf/2510.18212>

Type of AI	Definition	Example
Narrow AI (ANI)	Excels at <i>one</i> specific task.	ChatGPT, CLIP, facial recognition
General AI (AGI)	Adapts, reasons, and transfers knowledge across <i>all</i> domains.	Hypothetical (human-level)
Super AI (ASI)	Surpasses human capability in all intellectual tasks.	Theoretical

AGI = Artificial General Intelligence

Two views in AI community:

- A) skeptics (pessimistic on timeline/feasibility)
- B) optimists (positive on timeline/feasibility)

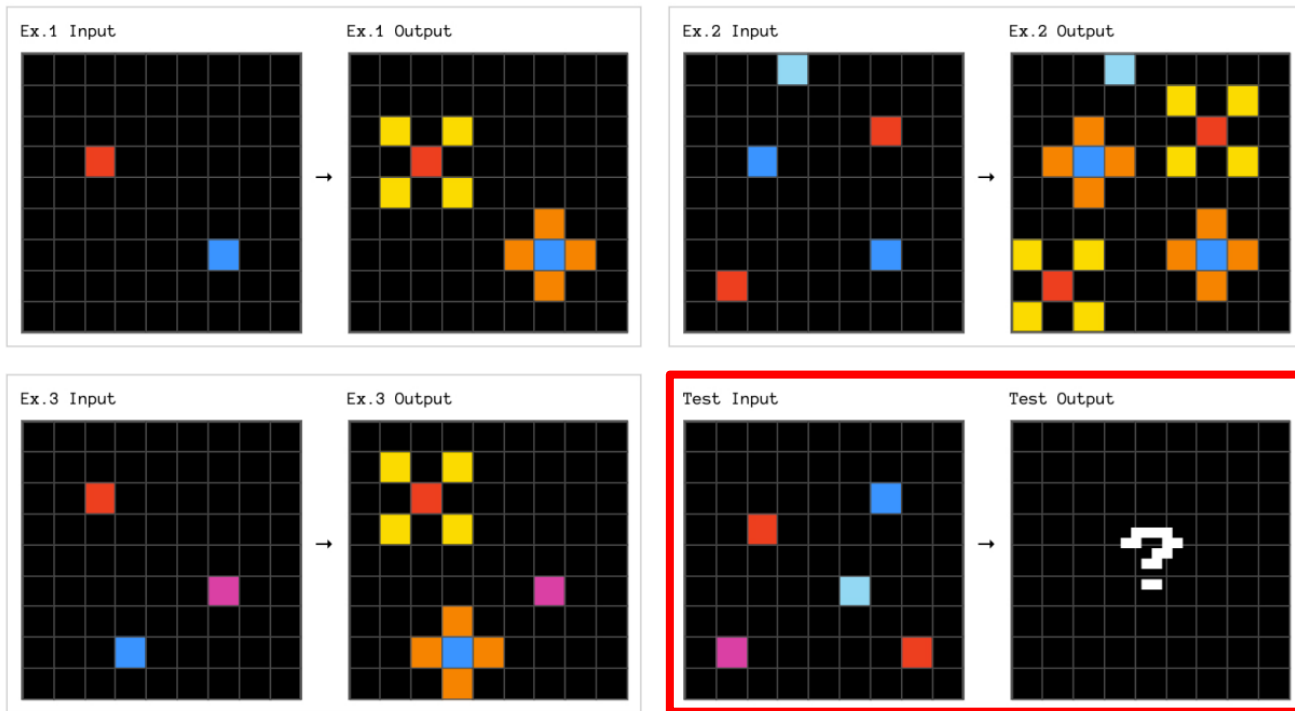
ARC-AGI "Abstract and Reasoning Corpus" for Artificial General Intelligence

AGI is a system that can efficiently acquire *new* skills outside of its training data.

ARC-AGI is the *only* AI benchmark that measures our progress towards *general* intelligence.

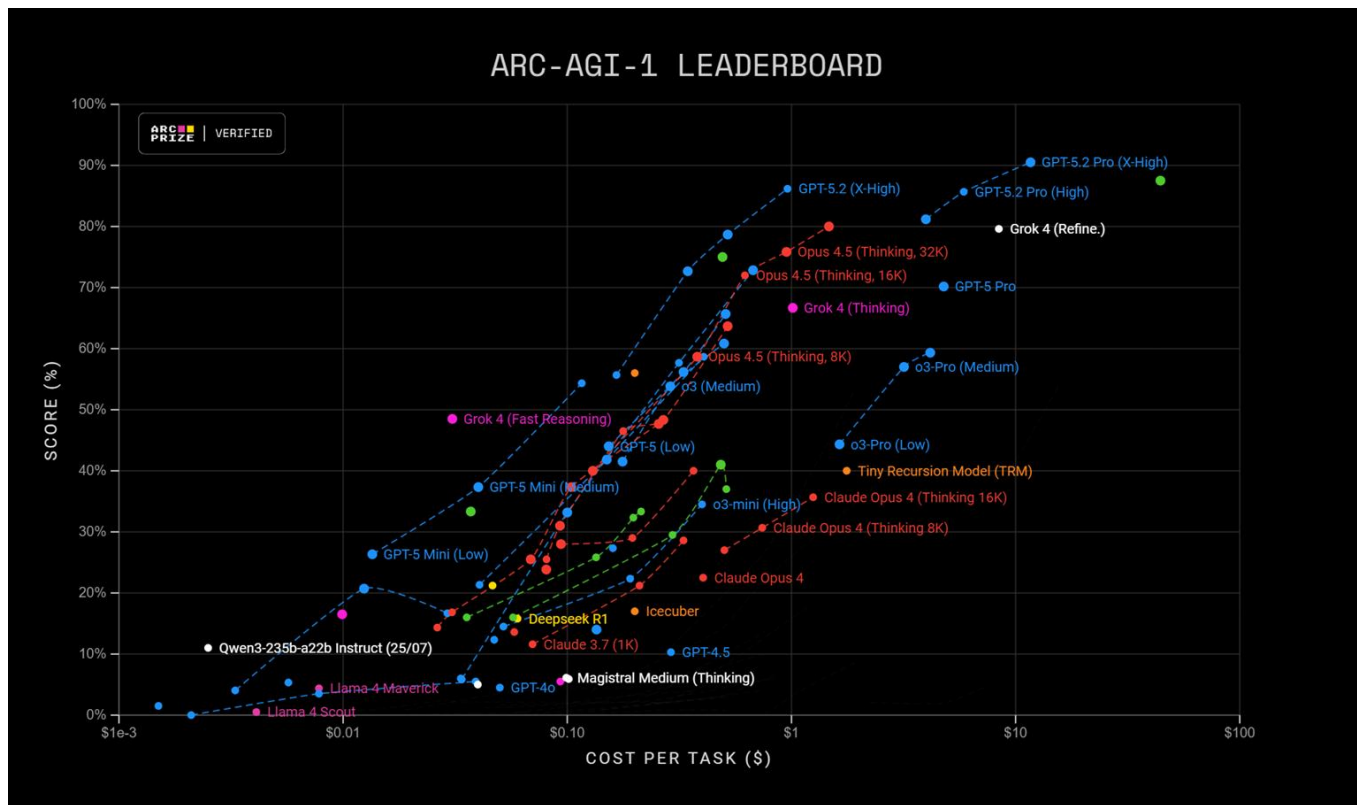
ARC-AGI "Abstract and Reasoning Corpus" for Artificial General Intelligence

Figure:
Example ARC-AGI task



ARC-AGI-1 Leaderboard

<https://arcprize.org/leaderboard>



ARC-AGI-2 Leaderboard

<https://arcprize.org/leaderboard>

